Answer Key — Competitive Exam Practice 1: Introduction to Operating Systems

GATE (Computer Science) Pattern

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Concepts That Click

Q1 (B). Translating C to assembly is the job of a *compiler* — a system program that runs in user mode — not the kernel. Scheduling (A), memory-frame allocation (C), and dispatching I/O to drivers (D) are all kernel responsibilities.

Q2 (C). Writing the base/limit registers is a *privileged* instruction; the processor traps it when the issuing process is in user mode. The registers are writable *by the kernel*, so (B) is false, and the instruction is not universally illegal, so (A) is false.

Q3 5. Fixed overhead = 3 μs. Block copy = 40 bytes/8 bytes = 5 units, at 0.4 μs each = 2 μs. Total = $3 + 2 = 5$ μs.

Q4 (B). In a monolithic kernel the driver shares the kernel's address space, so its memory corruption propagates into kernel state and can crash the system. In a microkernel the driver is a user-space server; a fault there is contained and the server can be restarted. (D) is wrong because microkernel drivers explicitly do *not* run in kernel mode.

Q5 (C). A system call is the controlled entry into the kernel for a privileged service. (A) is false: one POSIX call may map to several system calls. (B) is false: a system call executes in kernel mode. (D) is false: system calls routinely hand work to device drivers.

Q6 4098. The file is 8 MB = $8 \times 2^{20} = 8{,}388{,}608$ bytes; with a 4096-byte buffer there are 2048 buffers. Calls = 1 (open) + 2048 (read) + 2048 (write) + 1 (close) = 4098.

Q7 (B). Multiprogramming keeps several jobs in memory and switches on I/O waits to keep the CPU busy — its objective is utilization. Minimizing interactive response time (A) is the objective of *time-sharing*. Hard real-time guarantees (C) are a different class of system entirely.

Q8 (C). Presenting a clean, named abstraction (a file) over raw hardware (disk blocks) is precisely the OS acting as an *extended machine*. The resource-allocator (B) and control-program (A) views concern sharing and protection, not abstraction.